

AI Machine Learning

Reference questions - learning material AI Machine Learning



VIVES University of Applied Sciences
Bachelor in electronics-ICT



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Chapter 1 – What is Machine Learning?

- What is **Artificial Intelligence (AI)**?
- What is the relationship between **AI**, **Machine Learning**, and **Deep Learning**?
- What is the difference between an **AI model** and an **AI algorithm**?
- Machine Learning algorithms make predictions based on data, models, and “features.” Suppose you are asked to design an efficient spam filter using Machine Learning. Which “**features**” would you take into account, and explain why for each feature?

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Chapter 2 – Types of Machine Learning

- Discuss the essential differences between:
 - **Supervised** Machine Learning
 - **Unsupervised** Machine Learning
 - **Reinforcement** Learning
- What is the difference between a **regression model** and a **classification model**?
- What is the purpose of:
 - a **clustering** algorithm?
 - a **dimensionality reduction** algorithm?
 - a **generative algorithm**?
- Briefly discuss the technique of **dimensionality reduction** in unsupervised learning. Explain what **matrix factorization** and **singular value decomposition** are and where this technique is applied in practice.
- Discuss the principle of **reinforcement machine learning**.

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Chapter 3 – Linear regression

- Describe the **linear regression algorithm** in pseudocode.
- Briefly describe the characteristics of the **linear regression algorithm**, according to the following three methods:
 - Simple trick
 - Square trick
 - Absolute trick
- Why is the **mean absolute error** (MAE) or the **mean square error** (MSE) chosen when determining the error? Why is the **root mean square error** (RMSE) used in practice?
- How should a non-linear dataset be adjusted if the **linear regression** is transformed into **polynomial regression** using a fourth-degree polynomial?

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Chapter 4 – Underfitting, overfitting, testing and regularization

- How can underfitting, optimal fitting and overfitting be determined by comparing the **training error** and the **test error**?
- How can underfitting, optimal fitting and overfitting be determined by comparing the **validation error** and the **test error**?
- What are the **L1 norm** and the **L2 norm** for determining the complexity of a model?
- Discuss:
 - Lasso regression error
 - Ridge regression error

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Chapter 5 – The perceptron algorithm

- Why can the **step function** be used as an activation function in linear classification, such as in the perceptron algorithm? Why is the **step activation function** used less frequently in machine learning?
- Discuss how the **optimised perceptron algorithm** (= perceptron trick) works. Show how the learning process gradually determines the equation of an optimal straight line to split an original data set into two classes.
- Give a number of examples of a suitable **error function** for perceptron learning to support the learning process. Which error function do you propose and explain why this is a suitable error function for binary classification, for example?

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Chapter 6 – Logistic classifiers

- Logistic classifiers use the **sigmoid** as their activation function.
 - What is the mathematical form of the **sigmoid function**?
 - Draw the curve of the sigmoid for the values on the x-axis ranging from $-\infty$ to $+\infty$.
 - What are the advantages of using the sigmoid compared to a step function?
- Logistic classifiers use the **log loss error function**.
 - What is the mathematical form of the log loss error function?
 - Why is this error function preferred over the absolute or square error function?
- Discuss how the **optimised logistic regression algorithm** (= logistic trick) works. Show how the learning process gradually improves the classification of the original dataset.

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Chapter 7 – Accuracy of machine learning classification models

- What is meant by the “**accuracy**” of an ML model? Why is “accuracy” not sufficient to select an optimal model?
- **Confusion matrix:**
 - What is a confusion matrix?
 - How can you quickly determine the number of correctly and incorrectly classified points?
- **Metrics:**
 - What is the definition of the “recall” parameter and how is it determined using the confusion matrix?
 - What is the definition of the parameter “precision” and how is it determined using the confusion matrix?
 - What is the definition of the parameter “specificity” and how is it determined using the confusion matrix?
 - How is the F-score of a classification model determined?
- What is the **receiver operating characteristic** (ROC) and how does the **AUC** parameter help to optimise the **specificity** and **sensitivity** parameters?

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End of reference questions - learning material AI Machine Learning



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